Gabriel Pedde Ungureanu | Physics PhD | AI/ML Engineer

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Profile

PhD particle physicist (6 papers, ∼60 citations) with hands-on AI/ML engineering at EuroHPC scale: RLHF fine-tuning of Llama-3.2 (+11 pp on GSM8K), LLM multi-agent systems for scientific discovery, and distributed GPU training on Leonardo (CINECA). I build systems that pair LLM creativity with deterministic, verifiable ground truth.

Research & Engineering Experience

SISSA **Trieste, Italy**
Doctoral Researcher in Particle Physics and AI for Science *2022–2026*

- Non-perturbative QFT via symbolic computation on Ulysses (SISSA EuroHPC); 6 published papers, ∼60 citations.
- Building an LLM **"AI co-scientist"** (LangGraph, arXiv-MCP + RAG): multi-agent generation for General Relativity discovery, refereed by a **deterministic symbolic/numerical evaluator**—the LLM never grades itself.
- Recovers known vacuum metrics (Schwarzschild, Kerr, Taub–NUT) symbolically; Israel–Khan two-body solution to **machine precision** via self-improving Co-Scientist evolution.

Education

SISSA / ICTP Joint Programme **Trieste, Italy**
Master in High Performance Computing and AI (MHPC) *2025–2026*
Concurrent with the last year of PhD. Machine learning, deep learning, reinforcement learning, LLM agents, and distributed GPU computing.

Università Cattolica del Sacro Cuore **Brescia, Italy**
BSc Mathematics & MSc Physics (Laurea Triennale & Laurea Magistrale) *2017–2022*
Theoretical physics, QFT, condensed matter; strong analytical foundations. I did a master degree thesis in **quantum complexity theory** at the interface of holography and quantum information

Awards & Publications

6 peer-reviewed papers in theoretical physics (∼**60 citations**) — INSPIRE-HEP | Google Scholar

Best graduate of the year, MSc Physics — Università Cattolica del Sacro Cuore

110/110 *cum laude* — MSc Physics | 110/110 *cum laude* — BSc Mathematics

Istituto Toniolo merit fellowship — BSc, Università Cattolica del Sacro Cuore

Selected Projects — AI & ML

LLM RL fine-tuning: Implemented RLOO/PPO from scratch to fine-tune **Llama-3.2-1B** on GSM8K with a rule-based reward signal and KL regularisation; **+11 pp answer accuracy** (35 % → 46 %), format compliance from 51 % to 100 %. Distributed on **Leonardo (CINECA)**. Stack: PyTorch, TRL, SLURM | GitHub

Multi-Agent LLM System: Built a multi-agent debate framework (CrewAI, Ollama) across **5 Qwen2.5 sizes** (0.5B–14B), N = 4 agents, 10 rounds. Models ≥3B converge with $< 10^{-3}$ **variance at round 0** — a sharp capacity-threshold phase transition independent of training. Stack: CrewAI, Ollama, Python | GitHub

Distributed GPU: CUDA kernel development through to cuBLAS; distributed dense linear algebra with **MPI + CUDA**. Lattice Boltzmann on a 2048×2048 grid: ∼**120**× **GPU speedup** over CPU (0.75 s vs 91.5 s, 806 GB/s bandwidth). MPI + OpenMP atmospheric simulation: near-linear scaling 2 → 32 nodes (3087 s → 317 s) | GitHub

Technical Skills

AI / ML: PyTorch, Hugging Face Transformers & TRL, scikit-learn, cuML

LLM / Agents: RLHF, REINFORCE/RLOO, PPO, GRPO, fine-tuning, RAG, MCP, LangGraph, LangChain, CrewAI, Ollama, Claude Code

ML methods: CNNs, autoencoders, representation learning, dimensionality reduction, Bayesian inference, clustering

Programming: Python, C++, C, Fortran, Mathematica, LaTeX; SQL, SQLite

Scale / HPC: CUDA, cuBLAS, OpenACC, MPI, OpenMP, SLURM, PETSc/SLEPc

MLOps: Docker, Singularity/Apptainer, Git, HDF5/NetCDF

Clusters: Leonardo (CINECA), Ulysses (SISSA), Argo (ICTP)

Languages: Italian (native) | English (C1) | Romanian (A2)

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